PLS^2-Net: A Physics-Guided Liquid Spectral-Sequence Coarse-to-Fine Reconstruction Network for Remote Sensing Spectral Superresolution

Huijie Jia, Chenlin Wu, Min Huang, and Qibing Zhu

***Abstract*—Hyperspectral remote sensing imagery provides dense spectral measurements that support material identification and fine-grained classification, but it is expensive to acquire and often limited in spatial resolution. On the contrary, RGB imagery is low-cost and easy to capture with rich spatial details, yet its few channels cannot faithfully represent fine spectral signatures. Spectral superresolution (SSR), which reconstructs hyperspectral images from low-spectral observations, is therefore of practical importance but remains highly illposed: multiple high-dimensional spectra may correspond to the same RGB measurement, and narrowband absorption features or abrupt local spectral variations may not be distinguishable in RGB space, causing systematic bias in regions with similar appearance but different materials. Moreover, many existing SSR models cannot adapt their update behavior to input-dependent spectral dynamics, leading to oversmoothing. To address these issues, we propose PLS^2-Net, a physics-guided liquid spectral-sequence coarse-to-fine network for remote sensing SSR. PLS^2-Net first produces a stable and interpretable coarse spectral skeleton using a multiscale interpretable coarse spectral generation (MICSG) module. It then performs input-driven bidirectional spectral-sequence modeling with a bidirectional liquid spectral dynamics (BiLSDs) module, improving the representation of long-range cross-band dependencies and directional spectral evolution. Finally, a SRF-Based Physical Consistency Constraint Module (SPC) module together with a null-space detail refiner (NDR) enforces forward consistency and compensates for high-frequency details in unobservable directions, enhancing both physical fidelity and detail restoration. Extensive experiments on Pavia University (PaviaU), Chikusei, and Indian Pines demonstrate that PLS^2-Net achieves consistently superior reconstruction accuracy over state-of-the-art methods, particularly in spectral-shape preservation and error suppression.**

***Index Terms*—Hyperspectral image reconstruction, liquid neural network (LNN), physics-guided learning, spectral response function (SRF) consistency, spectral superresolution (SSR).**

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Huijie Jia, Chenlin Wu, and Qibing Zhu are with the Key Laboratory of Advanced Process Control for Light Industry, Ministry of Education, Jiangnan University, Wuxi 214122, China (e-mail: 7231905012@stu.jiangnan.edu.cn; 7241905016@stu.jiangnan.edu.cn; zhuqibing@jiangnan.edu.cn).

Min Huang is with the School of Automation and Intelligent Science (School of Internet of Things Engineering), Jiangnan University, Wuxi, Jiangsu 214122, China, and also with the Key Laboratory of Advanced Process Control for Light Industry, Ministry of Education, Jiangnan University, Wuxi 214122, China (e-mail: huangmin2004@jiangnan.edu.cn).

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I. Introduction

HYPERSPECTRAL remote sensing imagery provides rich spectral information and has been widely used in land-cover recognition and classification, offering clear advantages in distinguishing different surface materials and capturing fine spectral characteristics [1], [2], [3]. However, hyperspectral imaging typically relies on sophisticated optical systems and scanning-based acquisition, which increases cost and limits imaging efficiency. Moreover, improving spectral resolution often comes at the expense of spatial resolution: within the same scene area, capturing more spectral bands generally reduces the spatial information represented by each pixel, thereby weakening the depiction of fine spatial details. By contrast, RGB imagery is inexpensive and easy to acquire, and it often provides high spatial resolution with stable and clear texture/structure cues. Yet, its limited channels cannot adequately represent fine-grained material spectra, making it difficult to directly discriminate targets that appear visually similar but differ substantially in spectral properties. Consequently, reconstructing hyperspectral imagery from high-spatial-resolution RGB observations—namely spectral superresolution (SSR)—has become a key technique for jointly exploiting spectral and spatial information and is of considerable importance for remote sensing applications [4], [5], [6].

Fundamentally, SSR is a severely illposed inverse problem in the spectral domain: inferring a B-dimensional spectrum from a C-channel observation (typically $B \gg C$) inevitably suffers from information loss and nonuniqueness. More critically, real-world spectra often contain fine-scale structures such as narrowband abrupt changes, sharp local peaks, or absorption valleys (e.g., material absorption signatures). These structures do not necessarily yield distinguishable responses in RGB's limited channels. As a result, in regions where visual appearance is similar but material composition differs, SSR models tend to exhibit systematic reconstruction bias. This challenge stems from the identifiability issue induced by dimensionality reduction: multiple distinct hyperspectral signals may project to nearly identical RGB responses, meaning that reliable reconstruction must leverage spatial context, spectral priors, and physical constraints to narrow the feasible solution space.

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Existing SSR methods have progressed from model/prior-driven formulations [7], [8] to deep learning-based spectral–spatial mapping [10], [11], [12], and more recently toward physics-guided [13] and interpretable learning [9]. Despite steady improvements, three gaps remain central for remote sensing SSR under severe underdetermination. First, convolutional neural network (CNN)-based approaches mainly exploit local spectral–spatial correlations, which often limits their ability to preserve global spectral-shape consistency when long-range cross-band interactions dominate the reconstruction. Second, global modeling modules (e.g., attention or sequence architectures) can aggregate information across bands, yet their update behavior is typically not content-adaptive; when spectra contain sharp local transitions or narrow absorption structures, insufficiently adaptive updating can lead to oversmoothing in critical intervals and local segment mismatch across bands. Third, while spectral response function (SRF)-guided constraints provide a principled route to enforce observation consistency and improve interpretability, many methods incorporate SRF consistency only as a soft penalty, which may leave residual projection bias. More importantly, SRF-based dimensionality reduction induces inherent nonuniqueness: even if the projected spectrum matches the observation, null-space degrees of freedom remain unconstrained and can manifest as unstable spectral oscillations or nonphysical local distortions. These challenges motivate a reconstruction framework that combines an interpretable coarse representation, input-adaptive bidirectional spectral dynamics, and structured physical correction.

To address the above challenges, we propose PLS^2-Net, a physics-guided liquid spectral-sequence coarse-to-fine network for remote sensing SSR. The method decomposes reconstruction into four sequential stages: interpretable coarse spectral skeleton generation, spectral-sequence dynamical updating, physics-consistent correction, and null-space detail refinement. In particular, the multiscale interpretable coarse spectral generation (MICSG) module first constructs a stable coarse spectral skeleton through a low-dimensional spectral basis–coefficient representation, thereby constraining the feasible solution space and improving training stability. The bidirectional liquid spectral dynamics (BiLSDs) module then organizes the coefficient maps as an ordered spectral sequence and performs input-adaptive bidirectional dynamic updating using a liquid neural network (LNN) [15], enabling more effective modeling of long-range cross-band dependencies and directional spectral evolution while suppressing spectral-shape distortions around abrupt changes and absorption features. To further enforce physical fidelity, PLS^2-Net explicitly embeds the SRF forward model $\mathcal{R}(\mathbf{H}) = \mathbf{H}\mathbf{R}^\top$ into the reconstruction process: the spectral-response hard-consistency projection (SHP) performs differentiable hard-consistency correction to pull the intermediate estimate toward the SRF-feasible neighborhood, the prior-enhanced projection-consistent solver (PEPS) further introduces spectral-shape priors to alleviate nonuniqueness and spectral instability under severe underdetermination, and the null-space detail refiner (NDR) finally compensates high-frequency details under an explicit null-space constraint so that detail refinement is achieved without substantially disturbing the observable projection. In this way, PLS^2-Net improves reconstruction accuracy, spectral-shape fidelity, and physical consistency within a unified coarse-to-fine framework.

The main contributions of this work are summarized as follows.

1) We introduce the input-driven dynamic update mechanism of LNNs into SSR and develop BiLSD to bidirectionally model coefficient-domain spectral sequences, strengthening the modeling of long-range cross-band dependencies.
2) We propose a physics-guided coarse-to-fine framework integrating MICSG, SPC, and NDR to stabilize coarse spectral reconstruction, enforce SRF consistency, and recover details under null-space constraints.
3) Extensive experiments on Pavia University (PaviaU), Chikusei, and Indian Pines demonstrate that PLS^2-Net consistently outperforms representative methods in reconstruction accuracy, spectral fidelity, robustness, and downstream classification performance.

II. Related Work

Existing SSR methods can be broadly grouped into four categories according to their principal modeling mechanisms: model/prior-driven methods, CNN-based methods, sequence and global modeling methods, and physics-guided methods. This section reviews the representative architectures, underlying assumptions, and technical limitations of each category, and then clarifies the methodological differences between these approaches and the proposed PLS^2-Net.

A. Model/Prior-Driven SSR Methods

Early model-driven methods commonly relied on linear subspaces, sparse representations, or low-rank priors [16]. By constructing spectral dictionaries, low-dimensional manifolds, or structural regularizers, they attempted to invert low-spectral observations into hyperspectral signals with certain physical intuition and interpretability [17]. Yan et al. [18] proposed an SSR method based on dictionary optimization learning via spectral library alignment, which enhances the spectral resolution of multispectral images by incorporating a spectral library as a prior. Bu et al. [19] proposed a hyperspectral image superresolution method using transferable multiple subspace learning, which models the multisubspace structures of nonlocal patches via tensor low-rank representation. Xue et al. [20] proposed a hyperspectral image superresolution method based on spatial–spectral structured sparse low-rank representation, which integrates spatial and spectral subspace learning to improve fusion performance.

Nevertheless, such methods typically depend on strong prior assumptions and hand-crafted regularization, which limits their ability to capture complex nonlinear spectral–spatial coupling in real scenes. In addition, in regions that are visually similar yet spectrally distinct, they often underfit or produce oversmoothed spectra, making it difficult to recover fine spectral structures such as narrowband absorptions. These limitations

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motivated the shift toward more expressive data-driven models that can learn complicated spectral–spatial mappings from data.

B. CNN-Based SSR Methods

With the development of deep learning, CNNs have become a dominant paradigm for SSR [21]. By learning spectral–spatial mappings end-to-end, CNN-based models substantially improve reconstruction accuracy, particularly in scenes with rich textures or sharp structural boundaries [22]. Wu et al. [23] proposed the HPRN model, which integrates holistic priors (e.g., semantic information and deep features) and relation modules (e.g., semantic-driven spatial relation module) for SSR to regularize the solution space. Dixit and Yadav [24] proposed the U-SRN model, which employs a U-Net architecture for feature extraction and perceptual loss for single image superresolution, aiming to reduce network complexity and enhance visual quality. Farajzadeh et al. [25] investigated a hyperspectral image superresolution method using multiscale decomposition with NSCT and 3D-CNNs, optimized by similarity criteria for decomposition levels and network design.

However, the inherently local receptive field of CNNs makes them better suited for modeling local correlations, while their ability to capture long-range interband dependencies and global cross-band relationships remains limited. Meanwhile, under the illposed setting, training solely with reconstruction losses may induce nonphysical spectral fluctuations or erroneous absorption patterns in certain bands—effects that are especially pronounced for narrowband structures that RGB cannot reliably distinguish. Therefore, enhancing global cross-band dependence modeling has become an important direction for improving the reliability of SSR.

C. Sequence and Global Modeling for Cross-Band Dependence

To alleviate the deficiency in long-range dependence modeling, attention mechanisms and global sequence models such as LSTM/Transformers have been introduced into SSR to explicitly capture global relationships across space and across bands [26], [27]. Zhou et al. [28] proposed a hierarchical semantic-aware CNN-Transformer (HSACT) network for SSR of remote sensing images, utilizing an initial spectral estimation module and attention-driven mechanisms to improve spatial and spectral detail recovery. El-Gabri et al. [29] proposed a deep local residual attention network (DLRA-Net) with contextual refinement for SSR, integrating local low-rank spectral priors and global contextual information to enhance reconstruction accuracy. He et al. [30] proposed a dense spectral transformer (DsTer) integrated with ResNet for SSR of multispectral remote sensing images, leveraging transformers to capture long-range dependencies and dense connections for efficient feature exploitation.

Despite their progress, these methods still face two notable limitations. First, when spectral curves contain narrowband abrupt variations or absorption peaks, existing SSR approaches often cannot adaptively modulate the model's update strength in response to the input, which weakens the model's sensitivity to critical spectral changes and leads to the loss of fine details in reconstruction.

D. Physics-Guided SSR Methods

In recent years, physics-guided SSR has attracted increasing attention [31]. A representative strategy is to incorporate SRF-based forward consistency into the loss function or network design, enforcing that the reconstructed output matches the spectral response of the actual observation [32]. This direction offers a plausible route to improved interpretability and cross-sensor generalization. Peng et al. [33] introduced the W-SpectralFormer framework, which employs a frequency-assisted transformer-based network (FATSSR) and a spectral-aware loss function integrated with optical properties of water as physical constraints for weakly supervised SSR. Li et al. [34] proposed a multiscale hyperspectral recovery network (MHRN) that leverages the spatial-structure consistency between RGB and hyperspectral images and incorporates physical constraints (e.g., spectral response loss and neighboring band loss) for enhanced SSR. Fang et al. [35] developed the CS2DIPs method, which utilizes coupled spatial and spectral deep image priors (DIPs) integrated with nonnegative matrix–vector tensor factorization (NMVTF) and physical constraints (e.g., nonnegativity and sum-to-one) for unsupervised hyperspectral image superresolution.

However, a key contradiction remains: many existing works impose SRF consistency merely as a soft penalty, without a structured and explicit correction mechanism. As a consequence, the output may only "approximately" satisfy projection consistency during training, while still retaining noticeable bias in the observation domain—bias that can be amplified under domain shifts. Moreover, due to the severe nonidentifiability introduced by SRF-based dimensionality reduction, projection-consistent solutions are not unique. Without constraints on the null-space degrees of freedom, reconstructed spectra may become unstable or exhibit localized nonphysical oscillations.

III. Method

The overall architecture of PLS^2-Net is shown in Fig. 1. PLS^2-Net follows a coarse-to-fine reconstruction strategy. The four modules of PLS^2-Net address different sources of ambiguity in RGB-to-HSI reconstruction. MICSG stabilizes the coarse estimate through a low-dimensional and interpretable subspace representation. BiLSD operates on the coefficient-domain sequence rather than unconstrained full spectra, enabling stable modeling of long-range and directional spectral dependencies. SPC projects the intermediate estimate toward the SRF-feasible set and regularizes the remaining null-space freedom, while NDR refines details on top of this physically anchored baseline. Therefore, the overall architecture is an ordered decomposition of the illposed inverse problem rather than a heuristic combination of modules.

A. MICSG Module

To alleviate the illposedness of SSR and improve the interpretability of intermediate representations, we design a

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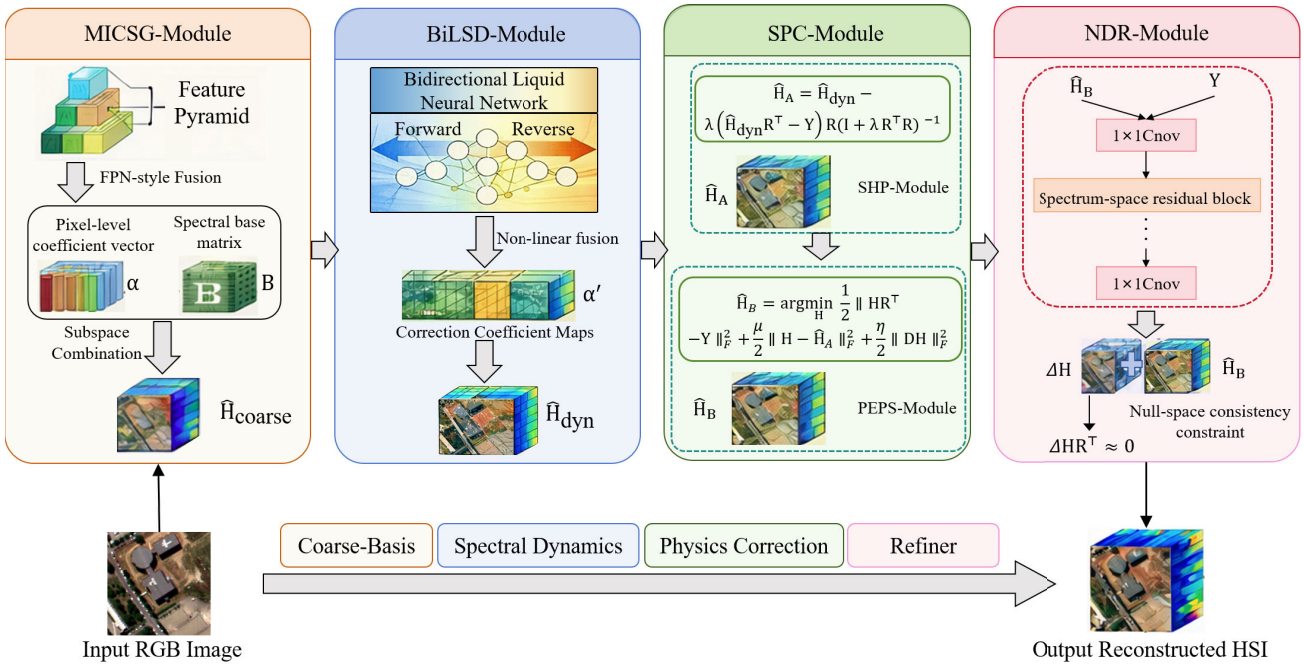


Fig. 1. Overall architecture of PLS^2-Net, consisting of the MICSG coarse spectral generation module, the BiLSDs module, the SPC SRF-consistent physics constraint module (SHP+PEPS), and the NDR module.

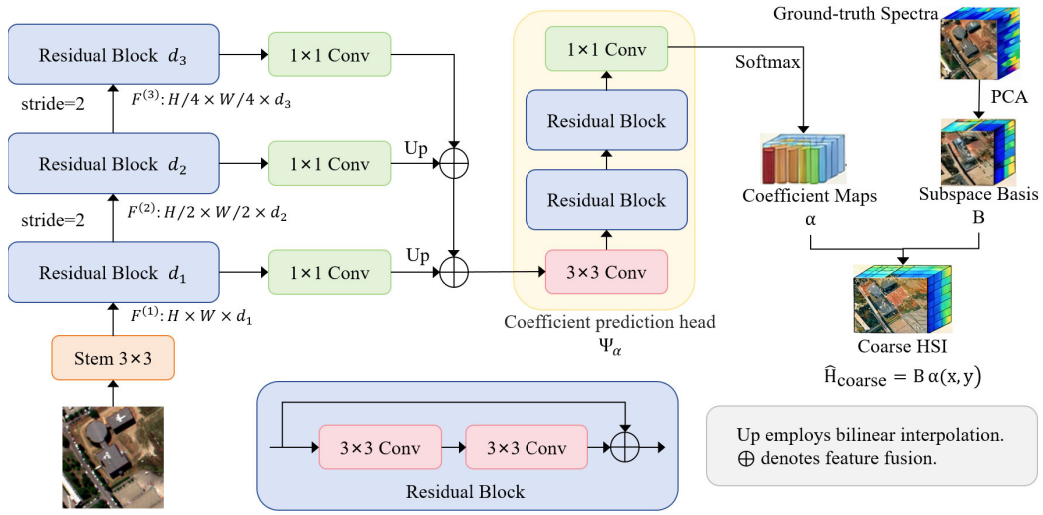


Fig. 2. Network framework of the MICSG module for interpretable coarse spectral skeleton generation.

MICSG module, as shown in Fig. 2. Given an RGB image $\mathbf{Y} \in \mathbb{R}^{H\times W\times 3}$, MICSG directly outputs a coarse hyperspectral estimate $\hat{\mathbf{H}}_{\text{coarse}} \in \mathbb{R}^{H\times W\times B}$. MICSG explicitly parameterizes the coarse reconstruction as a linear combination in a low-dimensional spectral subspace. In this way, the feasible solution space is constrained at the front end of the network, which mitigates spectral hallucination in unidentifiable regions and provides a stable coarse spectral skeleton for subsequent physics-consistent correction.

1) Multiscale Spectral–Spatial Representation Learning: MICSG aims to extract stable spatial structures and textures from the input RGB image $\mathbf{Y} \in \mathbb{R}^{H\times W\times 3}$. The input $\mathbf{Y}$ is first mapped to a base feature space by a shallow 3×3 convolutional stem. We then adopt a three-level pyramid encoder ($L = 3$) corresponding to $\times 1$, $\times 2$, and $\times 4$ scales. At each scale, lightweight residual blocks are used to refine features. Each residual block consists of two 3×3 convolutions with an intermediate nonlinearity and a residual addition

$$\mathrm{RB}(\mathbf{X}) = \mathbf{X} + \mathrm{Conv}_{3\times 3}\left(\sigma\left(\mathrm{Conv}_{3\times 3}(\mathbf{X})\right)\right) \tag{1}$$

where both $\mathrm{Conv}_{3\times 3}$ layers use stride = 1 and padding = 1 to preserve spatial resolution, and $\sigma(\cdot)$ denotes an activation function.

For scale progression, MICSG employs strided convolutions (stride = 2) to downsample features and obtain a set of multiscale feature maps

$$\{\mathbf{F}^{(l)}\}_{l=1}^{L} = \Phi_{\text{enc}}(\mathbf{Y}), \quad \mathbf{F}^{(l)} \in \mathbb{R}^{\frac{H}{s_l}\times\frac{W}{s_l}\times d_l} \tag{2}$$

where $s_l \in \{1, 2, 4\}$ denotes the scale factor and $\{d_l\}$ are the channel widths at different levels. Lower level features $\mathbf{F}^{(1)}$

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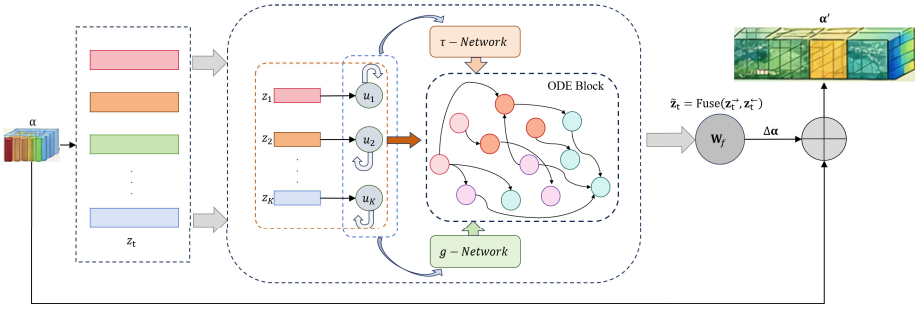


Fig. 3. Architecture of the BiLSDs module. LNN cell, where the input-driven τ_t-network and g-network dynamically generate the time constant and update gate to modulate continuous spectral state evolution. The state is updated through an ODE-inspired update rule. Two LNN streams propagate along the spectral sequence in opposite directions (low → high and high → low) to capture directional cross-band dependencies.

emphasize edges, textures, and local structures, whereas higher level features $\mathbf{F}^{(3)}$ capture global context and cross-region consistency.

To fully integrate multiscale information, we adopt an FPN-style top-down fusion strategy. In particular, features at each scale are first projected by lateral 1×1 convolutions to align channel dimensions to a unified width d. Then, higher level features are progressively upsampled and summed with lower level ones, followed by a 3×3 convolution for smoothing to suppress aliasing introduced by multiscale alignment. This yields a fused feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times d}$ at the original resolution. Such a design enables $\mathbf{F}$ to encode both fine local details and high-level semantics, providing a spectral–spatial representation foundation with structural coherence and contextual constraints for coarse spectral generation.

2) Interpretable Coarse Spectral Parameterization: Based on the fused feature map $\mathbf{F}$, a lightweight coefficient prediction head $\Psi_\alpha(\cdot)$ outputs unnormalized logits

$$\mathbf{Z} = \Psi_\alpha(\mathbf{F}), \quad \mathbf{Z} \in \mathbb{R}^{H \times W \times K} \tag{3}$$

where $\Psi_\alpha(\cdot)$ is composed of residual blocks and a 1×1 output layer, predicting a K-dimensional coefficient response for each pixel while preserving spatial structures. A Softmax is then applied to obtain constrained coefficient maps

$$\boldsymbol{\alpha}(x, y) = \text{Softmax}(\mathbf{Z}(x, y)) \in \mathbb{R}^K \tag{4}$$

which guarantees $\alpha_k(x, y) \geq 0$ and $\sum_{k=1}^{K} \alpha_k(x, y) = 1$.

During coarse spectral generation, MICSG explicitly adopts an interpretable subspace-basis–coefficient-map parameterization to reduce reconstruction freedom and suppress nonphysical solutions in unidentifiable regions. Let the subspace dimension be K ($K \ll B$). The coarse spectrum at each pixel (x, y) is synthesized by a linear combination of subspace bases

$$\hat{\mathbf{H}}_{\text{coarse}}(x, y, :) = \mathbf{B}\,\boldsymbol{\alpha}(x, y), \quad \hat{\mathbf{H}}_{\text{coarse}} \in \mathbb{R}^{H \times W \times B} \tag{5}$$

where $\mathbf{B} \in \mathbb{R}^{B \times K}$ is the spectral basis matrix. $\mathbf{B}$ is initialized using PCA statistics computed from the ground-truth spectra in the training set and is allowed to be slightly fine-tuned during training. This explicit subspace modeling restricts coarse reconstruction onto a low-dimensional spectral manifold, substantially shrinking the feasible solution space, stabilizing the coarse estimate, and reducing spurious oscillations in regions where the observation provides insufficient information.

B. BiLSDs Module

In SSR, beyond spatial features and local structures, long-range interband dependencies and global cross-band correlations are crucial for accurate reconstruction. To better capture such complex cross-band relationships, we introduce an LNN as the core component of the proposed BiLSDs module. Thanks to its input-driven mechanism, BiLSD can dynamically modulate both the frequency and strength of state updates according to spectral content, which is particularly beneficial when spectral curves contain abrupt local variations or narrow absorption features. BiLSD performs bidirectional propagation along the spectral index—forward (low → high) and backward (high → low)—to aggregate information across spectral groups, and then applies a residual correction to the coefficient map $\boldsymbol{\alpha}$, as shown in Fig. 3. This design enhances global spectral-shape consistency while suppressing local spectral oscillations.

1) LNN Architecture and Working Principle: An LNN achieves adaptive sequence modeling by introducing input-dependent time constants and update strengths, enabling flexible adjustment of the state-update rate at each step. Unlike conventional sequence models with fixed update patterns, an LNN automatically determines how information should propagate based on the current input, thereby avoiding rigid or "over-inertial" updates.

In general, the continuous state evolution of the LNN can be expressed as

$$\frac{d\mathbf{h}(t)}{dt} = f_\theta(\mathbf{h}(t), \mathbf{u}(t), t) \tag{6}$$

where $\mathbf{h}(t)$ is the hidden state, $\mathbf{u}(t)$ is the input spectral token, and $f_\theta(\cdot)$ denotes the learnable liquid dynamics. In BiLSD, the ordered spectral positions are treated as discrete steps of this continuous state evolution.

Formally, the state evolution can be written as

$$\mathbf{z}_{t+1} = \mathbf{z}_t + \Delta\mathbf{z}_t \tag{7}$$

where $\mathbf{z}_t$ denotes the state at step t, and $\Delta\mathbf{z}_t$ is the state increment. A basic form of the increment is

$$\Delta\mathbf{z}_t = \phi(\mathbf{W}_z\mathbf{z}_t + \mathbf{W}_u\mathbf{u}_t + \mathbf{b}) \tag{8}$$

where $\phi(\cdot)$ is a nonlinear activation function, $\mathbf{W}_z$ and $\mathbf{W}_u$ are learnable weight matrices, $\mathbf{u}_t$ denotes the spectral token (i.e., the embedding of the tth spectral position), and $\mathbf{b}$ is a bias

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term. This formulation enables the model to update its state adaptively according to the input-driven token $\mathbf{u}_t$.

To further enhance adaptivity, LNN introduces an input-driven time-constant term $\boldsymbol{\tau}_t$ and an update-gating term $\mathbf{g}_t$. Intuitively, $\boldsymbol{\tau}_t$ controls the effective memory length (how fast past information decays), while $\mathbf{g}_t$ controls the update magnitude (how strongly new information is injected). They are computed as

$$\boldsymbol{\tau}_t = \text{sigmoid}\left(\mathbf{W}_\tau \mathbf{u}_t + \mathbf{b}_\tau\right), \quad \mathbf{g}_t = \text{sigmoid}\left(\mathbf{W}_g \mathbf{u}_t + \mathbf{b}_g\right) \tag{9}$$

where $\mathbf{W}_\tau$ and $\mathbf{W}_g$ are learnable parameters and sigmoid$(\cdot)$ denotes the sigmoid function. With $\boldsymbol{\tau}_t$ and $\mathbf{g}_t$ being driven by $\mathbf{u}_t$, the network can adaptively adjust both information propagation length and update strength across different spectral regions, which is essential for modeling long-range dependencies.

2) BiLSD Architecture: BiLSD takes the coefficient map predicted by MICSG, $\boldsymbol{\alpha} \in \mathbb{R}^{H\times W\times K}$, and arranges it into a spectral sequence along the band index, where α_t denotes the coefficient channel at spectral position t. Each scalar coefficient $\alpha_t(x, y)$ is first embedded into a D-dimensional spectral token $\mathbf{u}_t(x, y)$ to drive the LNN updates.

BiLSD then performs forward and backward LNN propagation to capture both long-range cross-band dependencies and directional spectral evolution. For the forward stream, the state $\mathbf{z}_t^{\rightarrow}$ is updated as

$$\mathbf{z}_{t+1}^{\rightarrow} = \mathbf{z}_t^{\rightarrow} + \Delta t\left[-\boldsymbol{\tau}_t^{\rightarrow} \odot \mathbf{z}_t^{\rightarrow} + \mathbf{g}_t^{\rightarrow} \odot \phi\left(\mathbf{W}_z \mathbf{z}_t^{\rightarrow} + \mathbf{W}_u \mathbf{u}_t + \mathbf{b}\right)\right] \tag{10}$$

where $\phi(\cdot)$ is a nonlinear activation function, $\odot$ denotes elementwise multiplication, and $\boldsymbol{\tau}_t^{\rightarrow}$ and $\mathbf{g}_t^{\rightarrow}$ are the input-driven time-constant and gating terms obtained from $\mathbf{u}_t$ via lightweight mappings (as in Section III-B1). The backward stream $\mathbf{z}_t^{\leftarrow}$ follows the same form but is propagated in the reverse direction (high $\rightarrow$ low), enabling complementary contextual aggregation.

After obtaining the bidirectional states $\mathbf{z}_t^{\rightarrow}$ and $\mathbf{z}_t^{\leftarrow}$, BiLSD fuses them at each spectral position t to form a direction-enhanced representation

$$\tilde{\mathbf{z}}_t = \text{Fuse}\left(\mathbf{z}_t^{\rightarrow}, \mathbf{z}_t^{\leftarrow}\right) = \mathbf{W}_f\left[\mathbf{z}_t^{\rightarrow}; \mathbf{z}_t^{\leftarrow}\right] + \mathbf{b}_f \tag{11}$$

where $[\cdot;\cdot]$ denotes concatenation and $\mathbf{W}_f$ is a learnable linear fusion matrix.

BiLSD outputs a residual correction $\Delta\boldsymbol{\alpha} \in \mathbb{R}^{H\times W\times K}$ for the coefficients

$$\Delta\alpha_t(x, y) = \mathbf{w}_o^\top \tilde{\mathbf{z}}_t(x, y) + b_o, \quad \boldsymbol{\alpha}' = \boldsymbol{\alpha} + \Delta\boldsymbol{\alpha}. \tag{12}$$

Finally, the dynamically enhanced coarse spectrum is synthesized using the spectral basis $\mathbf{B}$

$$\hat{\mathbf{H}}_{\text{dyn}}(x, y, :) = \mathbf{B}\,\boldsymbol{\alpha}'(x, y) \tag{13}$$

where $\hat{\mathbf{H}}_{\text{dyn}}$ is then fed into the subsequent SRF-based physical consistency module for forward-consistent correction and prior-enhanced solving.

C. SRF-Based Physical Consistency Constraint Module (SPC)

To explicitly suppress spectral hallucinations induced by the illposed inverse nature of SSR, we propose an SPC. SPC leverages the known camera SRF to construct a forward imaging operator and performs a two-step structured correction on the coarse spectrum predicted by BiLSD. First, the SHP layer removes forward inconsistency residuals through a closed-form, differentiable update, pulling the estimate back toward the physically feasible neighborhood. Second, the PEPS injects an explicit spectral-shape prior while preserving forward consistency, thereby selecting a more physically plausible solution from the severely underdetermined feasible set.

Let the network input be $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$, where C is the number of observed channels, and let the hyperspectral band number be B. For a spatial position p, let $\hat{\mathbf{x}}_p \in \mathbb{R}^B$ denote the reconstructed hyperspectral spectrum and $\mathbf{y}_p \in \mathbb{R}^C$ denote the corresponding observed low-spectral vector. The SRF is represented by $\mathbf{R} \in \mathbb{R}^{C\times B}$, where each row describes the response of one observed channel over the B hyperspectral bands. For an RGB observation, $C = 3$. The physical forward projection is expressed as

$$\hat{\mathbf{y}}_p = \mathbf{R}\hat{\mathbf{x}}_p \tag{14}$$

where $\hat{\mathbf{y}}_p$ is the observation reconstructed from the estimated hyperspectral spectrum. Accordingly, physical consistency requires $\hat{\mathbf{y}}_p$ to be close to the actual observation $\mathbf{y}_p$. For the entire image, let $\hat{\mathbf{H}} \in \mathbb{R}^{H\times W\times B}$ denote the reconstructed hyperspectral image and $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$ denote the observed image. The corresponding tensor-form forward model is

$$\mathcal{R}\left(\hat{\mathbf{H}}\right) = \hat{\mathbf{H}}\mathbf{R}^\top. \tag{15}$$

This forward model provides the physical basis for both the structured correction in SPC and the SRF-consistency loss used during training.

1) Spectral-Response Hard-Consistency Projection (SHP) Layer: The SHP layer aims to reduce the observation inconsistency of the current estimate $\hat{\mathbf{H}}_{\text{dyn}}$ under the SRF forward model via an explicit projection update, i.e., enforcing that its forward projection under $\mathcal{R}(\mathbf{H}) = \mathbf{H}\mathbf{R}^\top$ is as close as possible to the observation $\mathbf{Y}$. In particular, SHP first computes the forward residual $\hat{\mathbf{H}}_{\text{dyn}}\mathbf{R}^\top - \mathbf{Y}$, then maps this residual back to the hyperspectral domain via back-projection through $\mathbf{R}$. A stabilization preconditioner $(\mathbf{I}+\lambda\mathbf{R}^\top\mathbf{R})^{-1}$ is further introduced to control the update scale and numerical stability, yielding a closed-form differentiable correction.

Equivalently, SHP computes $\hat{\mathbf{H}}_A$ as the solution of the following quadratic problem:

$$\hat{\mathbf{H}}_A = \arg\min_{\mathbf{H}} \frac{1}{2}\left\|\hat{\mathbf{H}}_{\text{dyn}}\mathbf{R}^\top - \mathbf{Y}\right\|_F^2 + \frac{\lambda}{2}\left\|\mathbf{H} - \hat{\mathbf{H}}_{\text{dyn}}\right\|_F^2 \tag{16}$$

where the first term enforces forward-projection consistency, and the second term penalizes overly large corrections to avoid numerical instability under underdetermination; $\lambda > 0$ controls the strength of this stabilization. In essence, SHP can be interpreted as: without significantly deviating from $\hat{\mathbf{H}}_{\text{dyn}}$, it rapidly pulls the estimate back toward the physically feasible

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neighborhood defined by the SRF imaging model, reducing the projection bias that may remain if SRF consistency is imposed only as a soft loss term.

In implementation, SHP is not treated as an iterative inner-loop optimizer, but as a one-shot differentiable correction layer. Since the SRF matrix $\mathbf{R}$ is fixed for a given imaging setting, the stabilization term associated with $(\mathbf{I} + \lambda \mathbf{R}^\top \mathbf{R})^{-1}$ is shared across all pixels and can be precomputed. Therefore, the additional cost of SHP mainly comes from low-dimensional matrix multiplications along the spectral dimension, rather than iterative optimization.

2) Prior-Enhanced Projection-Consistent Solver (PEPS): Although SHP effectively corrects forward residuals, the strong dimensionality reduction induced by $\mathbf{R}$ makes the solution to $\mathbf{H}\mathbf{R}^\top = \mathbf{Y}$ nonunique. In particular, along the null-space directions of the forward operator, the estimate may still exhibit nonphysical high-frequency spectral oscillations or local spectral-shape distortions—even when the projection already matches the observation. Without constraints on these unobservable degrees of freedom, the reconstruction can become spectrally unstable despite being projection consistent. To address this, PEPS introduces an explicit spectral-shape prior while maintaining physical consistency, thereby selecting a more reasonable solution within the feasible set.

Concretely, based on $\hat{\mathbf{H}}_A$, PEPS solves the following strictly convex regularized inversion:

$$\hat{\mathbf{H}}_B = \arg\min_{\mathbf{H}} \frac{1}{2} \|\mathcal{R}(\mathbf{H}) - \mathbf{Y}\|_F^2 + \frac{\mu}{2} \left\|\mathbf{H} - \hat{\mathbf{H}}_A\right\|_F^2 + \frac{\eta}{2} \|\mathbf{D}\mathbf{H}\|_F^2 \tag{17}$$

where μ controls the proximity to $\hat{\mathbf{H}}_A$, η controls the strength of the spectral prior, and $\mathbf{D}$ is a difference operator along the wavelength dimension. The term $\|\mathbf{D}\mathbf{H}\|_F^2$ suppresses nonphysical high-frequency oscillations in the spectral trajectories and thus encourages spectral continuity and global shape stability. The output $\hat{\mathbf{H}}_B$ serves as the physically consistent coarse spectrum produced by SPC.

Here, the first term enforces SRF forward consistency; the second term anchors the solution to the SHP-corrected estimate to prevent drift under underdetermination; and the third term acts as a spectral-shape prior that constrains the variation of unobservable null-space components. Importantly, the target of PEPS is explicit: near the SRF-constrained feasible neighborhood, it further restricts the admissible behaviors of null-space degrees of freedom, so that the final output is not only projection-correct but also spectrally plausible.

Similarly, PEPS is implemented as a regularized linear solver rather than a multistep iterative procedure. Its effective system matrix is of the form $\mathbf{R}^\top \mathbf{R} + \mu \mathbf{I} + \eta \mathbf{D}^\top \mathbf{D}$, where $\mu > 0$ and $\eta \geq 0$. The proximal term and the spectral prior improve numerical conditioning and suppress unstable null-space oscillations. As this matrix is also independent of spatial position for a fixed SRF, the corresponding computation remains low-dimensional and does not introduce an additional iterative convergence burden when processing large images.

Through the sequential SHP and PEPS corrections, SPC imposes a structured physical constraint that upgrades the network prediction from projection consistency to projection consistency with enhanced spectral-shape stability.

D. Null-Space Detail Refiner

After the spectral-sequence dynamical correction by BiLSD and the SRF-based physical consistency constraint enforced by SPC, the resulting $\hat{\mathbf{H}}_B$ already exhibits strong forward consistency and a reasonable global spectral shape. However, because the SRF projection introduces severe dimensionality reduction and thus nonidentifiability, $\hat{\mathbf{H}}_B$ can still be oversmoothed in fine spectral–spatial structures, such as edge-associated spectral–spatial coupling details and localized spectral-shape variations around narrow absorption features. To further enhance detail representation without violating forward consistency, we propose an NDR, which compensates unobservable detail components through residual refinement subject to a null-space constraint.

1) Residual Refinement Modeling: NDR takes $\hat{\mathbf{H}}_B$ and the network input $\mathbf{Y}$ as a joint input, and predicts a hyperspectral detail residual $\Delta\mathbf{H} \in \mathbb{R}^{H \times W \times B}$ using a lightweight spectral–spatial residual network

$$\Delta\mathbf{H} = G_\theta\left(\left[\hat{\mathbf{H}}_B, \mathbf{Y}\right]\right), \quad \hat{\mathbf{H}} = \hat{\mathbf{H}}_B + \Delta\mathbf{H} \tag{18}$$

where $[\cdot, \cdot]$ denotes channelwise concatenation and $G_\theta(\cdot)$ is the NDR network. To ensure that refinement primarily compensates for details rather than re-estimating the main spectral structure, G_θ adopts a lightweight residual design. In particular, a 1×1 convolution first aligns channels to obtain an initial feature $\mathbf{F}_0$, followed by a stack of spectral–spatial residual blocks. Each block is composed of a spatial 3×3 convolution and a channel-mixing 1×1 convolution in series. Finally, a 1×1 convolution maps features to a B-channel output $\Delta\mathbf{H}$.

2) Null-Space Definition and Soft-Constrained Refinement: Let $\mathbf{R} \in \mathbb{R}^{C \times B}$ denote the SRF matrix and let $r = \text{rank}(\mathbf{R})$. The null space of the SRF forward operator is defined as

$$\text{Null}(\mathbf{R}) = \left\{\mathbf{v} \in \mathbb{R}^B \mid \mathbf{R}\mathbf{v} = \mathbf{0}\right\}. \tag{19}$$

Let $\mathbf{N} \in \mathbb{R}^{B \times (B-r)}$ be an orthonormal basis matrix of Null($\mathbf{R}$). It therefore satisfies

$$\mathbf{R}\mathbf{N} = \mathbf{0}, \quad \mathbf{N}^\top \mathbf{N} = \mathbf{I}. \tag{20}$$

Under an ideal exact null-space refinement, the detail residual can be parameterized as

$$\Delta\mathbf{x}_p = \mathbf{N}\Delta\mathbf{z}_p \tag{21}$$

and the refined spectrum is

$$\hat{\mathbf{x}}_p = \hat{\mathbf{x}}_{c,p} + \mathbf{N}\Delta\mathbf{z}_p \tag{22}$$

where $\Delta\mathbf{z}_p \in \mathbb{R}^{B-r}$ denotes the coordinates of the detail residual in the null-space basis. Because $\mathbf{R}\mathbf{N} = \mathbf{0}$, this ideal parameterization gives

$$\mathbf{R}\hat{\mathbf{x}}_p = \mathbf{R}\hat{\mathbf{x}}_{c,p}. \tag{23}$$

In the proposed implementation, NDR directly predicts a B-dimensional residual $\Delta\mathbf{x}_p$ to retain flexible spectral–spatial

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detail modeling. To formally characterize its relationship with the null space, the predicted residual can be decomposed as

$$\Delta\mathbf{x}_p = \underbrace{\mathbf{N}\mathbf{N}^\top\Delta\mathbf{x}_p}_{\Delta\mathbf{x}_p^{\text{null}}} + \underbrace{\left(\mathbf{I}-\mathbf{N}\mathbf{N}^\top\right)\Delta\mathbf{x}_p}_{\Delta\mathbf{x}_p^{\perp}} \tag{24}$$

where $\Delta\mathbf{x}_p^{\text{null}}$ is the null-space component and $\Delta\mathbf{x}_p^{\perp}$ is the component outside the null space. The former can equivalently be written as

$$\Delta\mathbf{x}_p^{\text{null}} = \mathbf{N}\Delta\mathbf{z}_p, \quad \Delta\mathbf{z}_p = \mathbf{N}^\top\Delta\mathbf{x}_p. \tag{25}$$

Since $\mathbf{R}\Delta\mathbf{x}_p^{\text{null}} = \mathbf{0}$, only $\Delta\mathbf{x}_p^{\perp}$ affects the observable projection

$$\mathbf{R}\Delta\mathbf{x}_p = \mathbf{R}\Delta\mathbf{x}_p^{\perp}. \tag{26}$$

Therefore, minimizing the null-space consistency loss

$$\mathcal{L}_{\text{null}} = \left\|\mathbf{R}\Delta\mathbf{x}_p\right\|_1 \tag{27}$$

suppresses the observable component $\Delta\mathbf{x}_p^{\perp}$ and encourages the predicted residual to approach the ideal null-space form $\mathbf{N}\Delta\mathbf{z}_p$. Consequently, NDR implements a soft-constrained approximation to explicit null-space parameterization, allowing detail recovery while minimizing changes to the observation-domain projection.

E. Loss Function

The overall training objective consists of three types of constraints: reconstruction supervision, physical consistency, and null-space refinement. Let the final network output be $\hat{\mathbf{H}}$ (produced by NDR), the coarse-stage output be $\hat{\mathbf{H}}_{\text{coarse}}$, the observation be $\mathbf{Y}$, and the ground-truth hyperspectral image be $\mathbf{H}$.

1) Spectral Reconstruction Loss: To ensure overall fidelity in both spectral and spatial domains, we adopt an ℓ_1 reconstruction loss

$$\mathcal{L}_{\text{rec}} = \left\|\hat{\mathbf{H}}-\mathbf{H}\right\|_1. \tag{28}$$

In addition, to explicitly stabilize the coarse spectral skeleton, we introduce auxiliary supervision on the coarse output

$$\mathcal{L}_{\text{coarse}} = \left\|\hat{\mathbf{H}}_{\text{coarse}}-\mathbf{H}\right\|_1. \tag{29}$$

This term encourages MICSG to learn a coarse representation aligned with the ground truth (GT) in terms of low-frequency spectral shape and structured spectral content, providing a stable basis for subsequent coefficient-domain updating in BiLSD and physics-consistent correction in SPC, and reducing error accumulation and spectral hallucination.

2) SRF Forward-Consistency Loss: To explicitly enforce compliance with the SRF forward imaging model, we define the SRF physical consistency loss at each spatial position as

$$\mathcal{L}_{\text{SPC}} = \left\|\mathbf{R}\hat{\mathbf{x}}_p-\mathbf{y}_p\right\|_1. \tag{30}$$

Its equivalent image-level form is

$$\mathcal{L}_{\text{SPC}} = \left\|\hat{\mathbf{H}}\mathbf{R}^\top-\mathbf{Y}\right\|_1. \tag{31}$$

This loss directly measures the discrepancy between the observed low-spectral image and the forward projection of the reconstructed hyperspectral image. It complements the structured physical correction performed by SPC: the SHP and PEPS layers impose physical consistency within the network architecture, whereas $\mathcal{L}_{\text{SPC}}$ enforces the same forward imaging relationship at the objective level.

3) Null-Space Refinement Loss: To ensure that NDR does not violate forward consistency during detail compensation, we impose a null-space constraint on $\Delta\mathbf{H}$

$$\mathcal{L}_{\text{null}} = \left\|\Delta\mathbf{H}\mathbf{R}^\top\right\|_1. \tag{32}$$

This term encourages the refinement residual to lie close to the null space of the forward projection operator, thereby compensating for details while minimizing its disturbance to the observable projection $\hat{\mathbf{H}}\mathbf{R}^\top$.

4) Total Loss: The total loss is given by a weighted combination

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda_c\mathcal{L}_{\text{coarse}} + \lambda_p\mathcal{L}_{\text{SPC}} + \lambda_n\mathcal{L}_{\text{null}} \tag{33}$$

where λ_c, λ_p, and λ_n are weighting coefficients that balance reconstruction accuracy, coarse-skeleton stability, physical consistency, and null-space refinement.

IV. Experimental Results and Analysis

A. Datasets

We evaluate the proposed method on three widely used remote sensing hyperspectral datasets: PaviaU [36], Chikusei [37], and Indian Pines [38]. In our SSR setting, the original HSI serves as ground-truth supervision, and the paired RGB observations are synthesized pixelwise using the known SRF.

PaviaU (ROSIS) has a spatial size of 610 × 340, a ground sampling distance of 1.3 m, and 103 bands covering 430–860 nm (VIS–NIR). Its clear urban edges and textures make it suitable for evaluating spatial detail preservation and spectral–spatial coupling recovery.

Chikusei (airborne VNIR) is 2517 × 2335 with 2.5 m GSD, containing 128 bands over 363–1018 nm. With large-area coverage, diverse land-cover types, and strong scale variations, it is well suited for testing robustness in complex scenes.

Indian Pines (AVIRIS) is 145 × 145 with 20-m spatial resolution; the commonly used corrected version retains 200 bands after removing water-absorption/low-SNR bands, spanning 400–2500 nm (VIS–SWIR).

B. Implementation Details

All experiments are conducted on the same hardware platform: an Intel[1] Core[2] i9-10900X CPU, two NVIDIA GeForce RTX 4090 GPUs, and 64 GB RAM.

For PLS^2-Net, the main hyperparameters are set as follows: the subspace dimension is $K = 8$; the consistency strength in SHP is $\lambda = 0.1$. In PEPS, the weights of the data-consistency term, proximal term, and spectral prior term are set to $\mu = 1.0$ and $\eta = 0.01$, respectively. The loss weights are $\lambda_c = 0.1$, $\lambda_p = 0.5$, and $\lambda_n = 0.2$. All the hyperparameters will be further verified in the subsequent experimental analysis.

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TABLE I
QUANTITATIVE RESULTS FOR THE PAVIAU DATASET. BOLD TEXT INDICATES THE OPTIMAL METRIC, WHILE UNDERLINED TEXT INDICATES THE SECOND-BEST METRIC

Metric	J-SLoL	MHRN	HSCNN+	AWAN	LTRN	MST++	HSACT	PLS^2-Net
PSNR ↑	28.982	34.811	34.697	35.019	35.241	<u>38.879</u>	38.451	**39.899**
SSIM ↑	0.8222	0.9076	0.9116	0.9185	0.9166	0.9547	<u>0.9608</u>	**0.9708**
ERGAS ↓	15.5187	7.7325	7.1640	7.2293	7.3864	6.5624	<u>6.5451</u>	**6.2064**
SAM ↓	8.9547	5.4628	5.5677	5.2314	5.4980	4.3325	<u>4.0319</u>	**3.4178**

During training, we use fixed-size patches as inputs. The initial learning rate is set to 2×10^{-4}, with a weight decay of 1×10^{-4}. The model is trained for 200 epochs with a batch size of 16.

C. Evaluation Metrics

To comprehensively evaluate hyperspectral reconstruction performance, we employ four widely used metrics: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), erreur relative globale adimensionnelle de synthése (ERGAS), and spectral angle mapper (SAM). PSNR measures the overall pixelwise reconstruction fidelity, while SSIM evaluates the preservation of spatial structures and local image characteristics. ERGAS reflects the normalized reconstruction error across spectral bands, and SAM measures the angular deviation between the reconstructed and ground-truth spectral vectors, thereby evaluating spectral-shape consistency. Higher PSNR and SSIM values indicate better performance, whereas lower ERGAS and SAM values are preferred. All compared methods are evaluated using the same metric implementation and evaluation protocol.

D. Results Analysis

To demonstrate the effectiveness of PLS^2-Net, we compare it with seven representative SSR methods, including J-SLoL [39], MHRN [34], HSCNN+ [41], AWAN [42], LTRN [43], MST++ [44], and HSACT [28].

Joint superresolution and low-rank learning (J-SLoL) combines sparse and low-rank learning to recover hyperspectral signals from low-spectral observations.

MHRN performs multiscale spectral reconstruction while exploiting RGB–HSI spatial-structure and imaging-consistency constraints.

Hyperspectral CNN+ (HSCNN+) enhances CNN-based hyperspectral recovery from RGB images via advanced residual/dense designs.

Attention weighted aggregation network (AWAN) introduces adaptive weighted attention to integrate features and improve spectral reconstruction from RGB images.

Low-rank tensor reconstruction network (LTRN) leverages a deep low-rank tensor representation for SSR.

Multistage spectralwise transformer (MST++) is a Transformer-based multistage model designed for efficient spectral reconstruction.

HSACT combines hierarchical convolutional feature extraction with Transformer-based global interaction for remote sensing RGB-to-HSI SSR.

These methods cover model-driven sparse and low-rank reconstruction, CNN-based spectral recovery, camera-sensitivity-aware reconstruction, low-rank tensor modeling, Transformer-based cross-band interaction, hierarchical CNN-Transformer modeling, and RGB-to-HSI imaging-consistency-guided reconstruction. For all compared methods, we use publicly available official implementations whenever possible and follow the training/testing protocol recommended in their papers. To ensure a fair comparison, all methods are trained and evaluated on the same dataset splits and the same paired RGB–HSI setting. Unless otherwise specified by the original implementations, we keep the input patch size, training epochs, batch size, optimizer, and learning-rate schedule consistent across methods, and report results under the same evaluation code and metrics.

Tables I–III summarize the quantitative comparisons between PLS^2-Net and state-of-the-art methods on the three datasets.

1) Pavia University: On PaviaU, PLS^2-Net achieves the best results of 39.899 dB / 0.9708 / 6.2064 / 3.4178 (PSNR/SSIM/ERGAS/SAM), outperforming strong competitors such as MST++ and HSACT. This indicates not only higher reconstruction accuracy but also better spectral-shape consistency. The RMSE maps in Fig. 4 further show that PLS^2-Net produces larger low-error regions, especially around material boundaries and sharp texture transitions, demonstrating more stable bandwise recovery in challenging urban structures.

2) Chikusei: On Chikusei, PLS^2-Net again achieves the best performance across all four metrics, with 42.848 dB PSNR, 0.9830 SSIM, 13.4146 ERGAS, and 1.8052 SAM. These gains indicate strong reconstruction ability in large scenes with complex textures and scale variations. As shown in Fig. 5, competing methods tend to suffer from local bias and detail attenuation in texture-rich regions, whereas PLS^2-Net better preserves fine structures and reduces local spectral distortions.

3) Indian Pines: On Indian Pines, PLS^2-Net achieves 48.343 dB PSNR, 0.9880 SSIM, 3.0236 ERGAS, and 1.2608 SAM, with the most notable advantage in SAM. Since this dataset contains many agricultural classes with similar spatial appearance but distinct spectral characteristics, SAM is particularly indicative of spectral-shape fidelity. The RMSE maps in Fig. 6 confirm that PLS^2-Net produces a lower and more uniform error background while significantly suppressing boundary-related high-error regions, highlighting its advantage in preserving material-specific spectral structures.

E. Spectral Curve Comparison

Fig. 7 presents spectral profile comparisons at selected regions/pixels on the three datasets. In particular, the plotted

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TABLE II

QUANTITATIVE RESULTS FOR THE CHIKUSEI DATASET. BOLD TEXT INDICATES OPTIMAL METRICS, WHILE UNDERLINED TEXT DENOTES SUBOPTIMAL METRICS

Metric	J-SLoL	MHRN	HSCNN+	AWAN	LTRN	MST++	HSACT	PLS^2-Net
PSNR ↑	37.914	40.482	40.936	41.649	41.002	<u>41.825</u>	41.216	**42.848**
SSIM ↑	0.9265	0.9496	0.9500	0.9605	0.9681	<u>0.9753</u>	0.9740	**0.9830**
ERGAS ↓	25.1280	20.1043	19.2563	16.4014	17.6444	15.9917	<u>14.6018</u>	**13.4146**
SAM ↓	5.3688	4.1092	4.2580	2.2770	2.4803	2.2581	<u>2.1057</u>	**1.8052**

TABLE III

QUANTITATIVE RESULTS FOR THE INDIAN PINES DATASET. BOLD TEXT INDICATES OPTIMAL METRICS, WHILE UNDERLINED TEXT DENOTES SUBOPTIMAL METRICS

Metric	J-SLoL	MHRN	HSCNN+	AWAN	LTRN	MST++	HSACT	PLS^2-Net
PSNR ↑	40.951	46.741	46.237	46.932	46.924	<u>47.859</u>	47.531	**48.343**
SSIM ↑	0.9362	0.9688	0.9620	0.9665	0.9635	0.9691	<u>0.9697</u>	**0.9880**
ERGAS ↓	4.5597	4.1072	3.7437	3.4200	3.6134	<u>3.1397</u>	3.3748	**3.0236**
SAM ↓	2.4510	2.2249	2.0484	2.0456	2.1041	<u>1.8819</u>	1.9316	**1.2608**

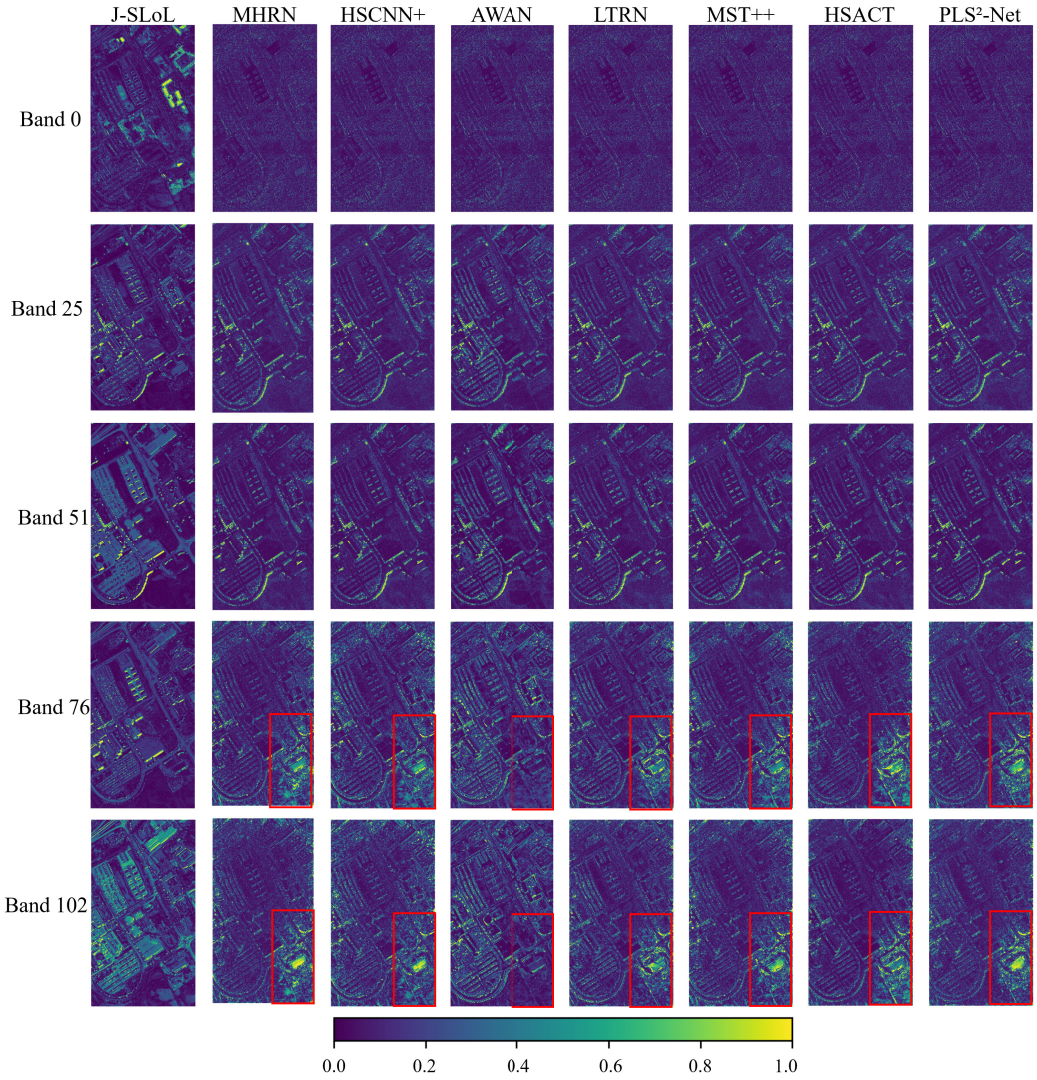


Fig. 4. Visualization of RMSE results for the J-SLoL, HRNet, HSCNN+, AWAN, LTRN, MST++, HSRNet, and PLS^2-Net methods on the PaviaU dataset across five bands: 0, 25, 51, 76, and 102.

bands are 0, 25, 51, 76, and 102 for PaviaU, 0, 31, 63, 95, and 127 for Chikusei, and 0, 54, 109, 164, and 219 for Indian Pines. Overall, several competing methods exhibit two typical types of errors around critical spectral transition intervals. The first is oversmoothing of local band responses, which leads to underestimated slopes in turning regions. The second is systematic bias at high wavelengths or near absorption-related variations, manifested as global upward/downward shifts or segmentwise deviations from the GT. On the contrary, PLS^2-Net consistently produces spectral curves that more

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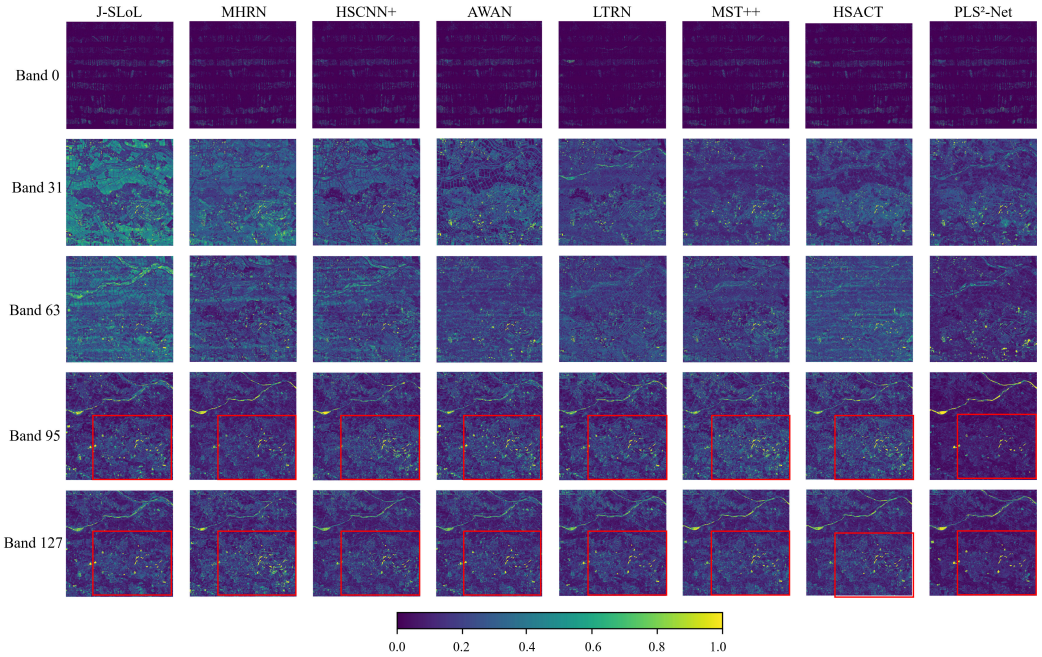


Fig. 5. Visualization of RMSE for J-SLoL, HRNet, HSCNN+, AWAN, LTRN, MST++, HSRNet, and PLS^2-Net methods across five bands (0, 31, 63, 95, 127) on the Chikusei dataset.

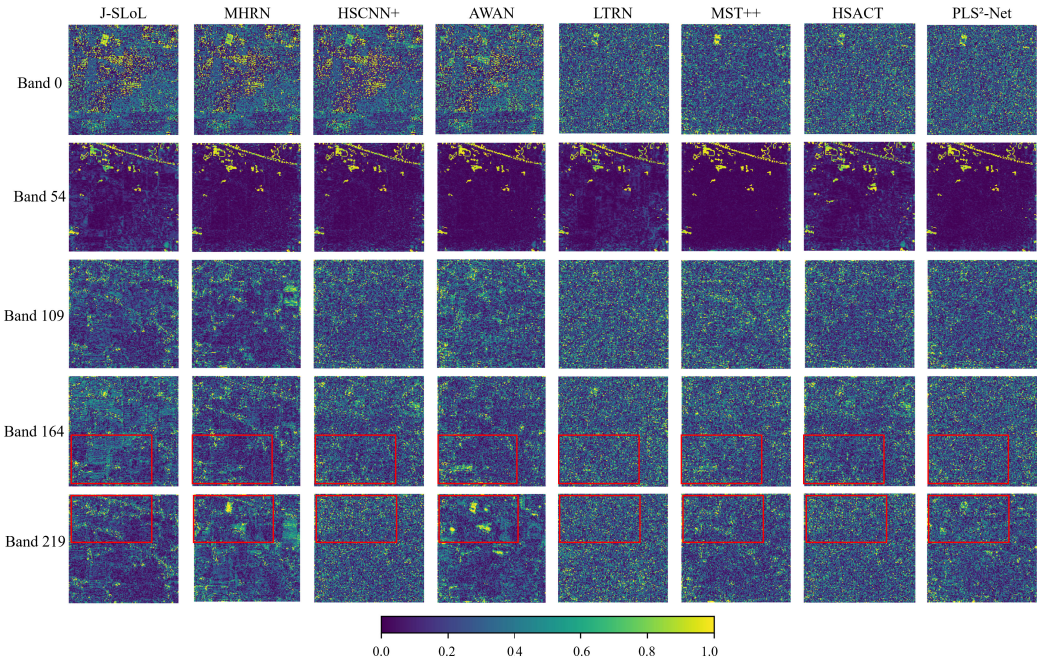


Fig. 6. Visualization of RMSE results for J-SLoL, HRNet, HSCNN+, AWAN, LTRN, MST++, HSRNet, and PLS^2-Net methods across five bands (0, 54, 109, 164, 219) on the Indian Pines dataset.

closely follow the ground-truth profiles across all three datasets. In particular, within the zoomed-in segments, PLS^2-Net maintains more consistent trends and amplitude levels, indicating that it is less prone to local segment mismatch during cross-band information propagation.

F. Model Complexity Analysis

As reported in Table IV, PLS^2-Net contains 13.5 M parameters and requires 5.9 G FLOPs. Compared with relatively high-complexity models such as MHRN (22.4 M parameters, 9.4 G FLOPs) and MST++ (15.4 M parameters, 6.0 G FLOPs), PLS^2-Net achieves a comparable level of modeling capacity while maintaining lower parameter counts and computational costs, suggesting that the proposed design and physics-guided constraints improve efficiency without incurring substantial overhead. In addition, lightweight methods such as J-SLoL (5.8 M, 2.6G FLOPs) and AWAN (4.6 M, 2.1G FLOPs) are computationally cheaper and thus more suitable for resource-constrained settings; however, their limited

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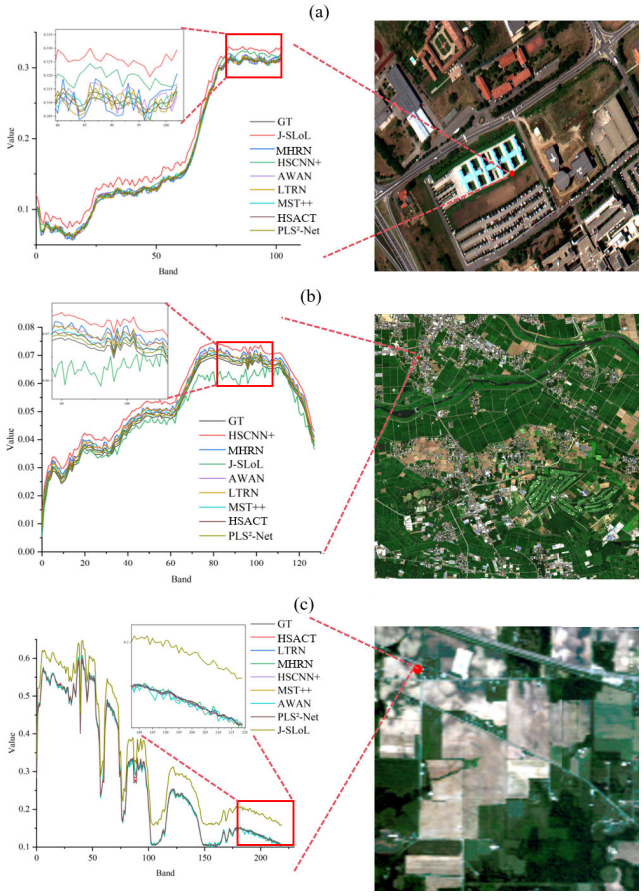


Fig. 7. Comparison of spectral curves between the J-SLoL, HRNet, HSCNN+, AWAN, LTRN, MST++, HSRNet, and PLS^2-Net methods and GT on (a) PaviaU, (b) Chikusei, and (c) Indian Pines datasets.

TABLE IV
COMPARISON OF MODEL COMPLEXITY ACROSS DIFFERENT METHODS

Method	Parameters (Millions)	FLOPs (Billions)
J-SLoL	5.8	2.6
MHRN	22.4	9.4
HSCNN+	14.1	6.5
AWAN	4.6	2.1
LTRN	12.3	4.8
MST++	15.4	6.0
HSACT	8.4	6.2
PLS^2-Net	13.5	5.9

representational power typically prevents them from achieving reconstruction quality comparable to PLS^2-Net, especially in terms of detail recovery and spectral-shape fidelity.

To further examine the tradeoff between complexity and performance, we visualize SAM, SSIM, Parameters, and FLOPs for all methods on the Chikusei dataset, as shown in Fig. 8. Although PLS^2-Net requires nontrivial computation (reflected by the marker size and color that encode parameters and FLOPs), its position in the SAM–SSIM plane indicates a clear performance advantage, particularly in achieving higher SSIM and lower SAM. By comparison, other high-complexity models such as MST++ incur higher computational costs but do not provide commensurate gains in SSIM and SAM. These results suggest that PLS^2-Net achieves a more favorable balance between reconstruction accuracy and

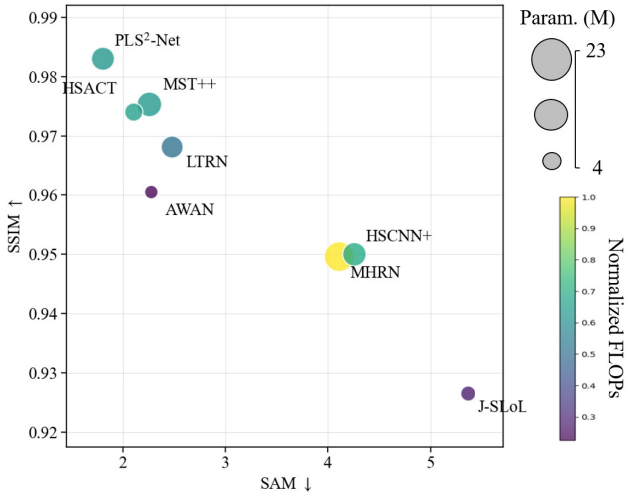


Fig. 8. Comparison of computational complexity and reconstruction accuracy among different methods on the Chikusei dataset. Bubble size represents the number of parameters, while bubble color denotes the normalized FLOPs.

computational burden, delivering larger performance improvements even when compared against models with similarly high—or higher—complexity.

G. Generalization Under Sensor-Response Perturbations

To further evaluate the stability of PLS^2-Net under nonideal sensing conditions, we additionally conduct controlled SRF-perturbation experiments beyond the nominal-SRF setting. In particular, the RGB observations are still synthesized using the nominal SRF during training, whereas at test time we perturb the center wavelengths of the SRF to simulate sensor calibration errors or unknown response offsets. Let the perturbation magnitude be δ. We consider four settings, namely $\delta = 0, \pm 5, \pm 10$, and ± 15 nm, where $\delta = 0$ corresponds to the standard evaluation protocol without mismatch.

As shown in Table V, as the SRF-perturbation magnitude gradually increases, PSNR and SSIM decrease mildly on all three datasets, while ERGAS and SAM increase accordingly. This indicates that mismatch in the forward response predictably affects reconstruction quality. Nevertheless, under mild-to-moderate perturbations (± 5 and ± 10 nm), the degradation of PLS^2-Net remains relatively limited, suggesting that the proposed framework is reasonably tolerant to small SRF calibration errors. Even under the stronger perturbation of ± 15 nm, the model still maintains comparatively stable reconstruction performance. This suggests that the interpretable coarse spectral skeleton from MICSG, the dynamic cross-band modeling of BiLSD, and the physics-consistent correction provided by SPC+NDR jointly help suppress spectral-shape drift and local instability caused by response mismatch.

H. Ablation Studies

1) Ablation of Network Components: To quantify the contribution of each core component, we conduct module-level ablations on PaviaU. The full PLS^2-Net is used as the reference, and one component is removed in each variant. Since MICSG and BiLSD are, respectively, responsible for

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TABLE V

ROBUSTNESS OF PLS^2-NET UNDER CONTROLLED SRF PERTURBATIONS

SRF perturbation	PaviaU				Chikusei				Indian Pines			
	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓
$\delta = 0$	39.899	0.971	6.206	3.418	42.848	0.983	13.415	1.805	48.343	0.988	3.024	1.261
$\delta = \pm 5$	39.421	0.965	6.355	3.566	42.353	0.981	13.906	1.915	48.072	0.982	3.118	1.338
$\delta = \pm 10$	38.818	0.960	6.599	3.781	42.013	0.977	14.283	2.071	47.658	0.975	3.254	1.452
$\delta = \pm 15$	38.141	0.958	6.933	4.049	41.424	0.972	14.721	2.255	47.123	0.968	3.436	1.912

TABLE VI

RESULTS OF ABLATION EXPERIMENTS FOR DIFFERENT MODULE COMPONENTS ON THE PAVIAU DATASET

Variant (PaviaU)	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓
Full (PLS^2-Net)	**39.899**	**0.9708**	**6.2064**	**3.4178**
w/o MICSG	38.972	0.9631	6.8449	3.9825
w/o BiLSD	39.284	0.9664	6.5382	3.6540
w/o SPC	39.062	0.9647	6.7921	3.9286
w/o NDR	39.521	0.9689	6.3368	3.5069

TABLE VII

COMPARISON OF DIFFERENT SPECTRAL-SEQUENCE MODELING MODULES ON THE PAVIAU DATASET

Sequence module	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓	Params (M)
BiLSTM	38.772	0.9619	6.8217	3.6716	13.6
BiGRU	38.938	0.9625	6.8874	3.6688	13.5
Transformer	39.284	0.9664	6.5382	3.6540	13.6
BiLSD (ours)	39.899	0.9708	6.2064	3.4178	13.5

interpretable coarse-skeleton generation and spectral-sequence modeling, structure-equivalent replacements are used: MICSG is replaced by a conventional CNN-based coarse branch, and BiLSD is replaced by a Transformer-style cross-band modeling module. For the physics-guided part, SPC and NDR are directly removed without substitution.

The Transformer replacement consists of two encoder blocks with embedding dimension D = 64, 4 attention heads, and FFN dimension 128. It takes the same coefficient-domain sequence as input and uses the same residual readout for $\Delta\alpha$, so that the comparison mainly reflects the difference in sequence-modeling mechanism rather than interface or scale mismatch.

As illustrated in Table VI, removing MICSG causes the most evident degradation in ERGAS and SAM, confirming the importance of an interpretable low-dimensional coarse spectral skeleton for stabilizing the solution space. Replacing BiLSD with Transformer also degrades all metrics, especially ERGAS and SAM, indicating that the liquid-state update is more effective than static global attention for maintaining spectral consistency. Removing SPC leads to clear deterioration, showing that SRF-consistent correction is critical for suppressing projection bias. Removing NDR mainly affects PSNR and SSIM, suggesting that it primarily improves fine details on top of the physically corrected baseline.

2) Comparison With Alternative Spectral-Sequence Models: To further validate the advantage of BiLSD in spectral-sequence modeling, we compare it with several representative alternatives on the PaviaU dataset, including BiLSTM, BiGRU, and Transformer. All the compared modules are inserted at the same position, operate on the same coefficient-domain sequence, and use aligned settings so that the comparison mainly reflects the difference in sequence-modeling mechanisms.

As illustrated in Table VII, under closely matched parameter budgets and FLOPs, BiLSD achieves the best performance across all four metrics. Compared with BiLSTM and BiGRU, it improves not only overall reconstruction accuracy but also spectral-shape consistency. Its advantage over Transformer is more pronounced, suggesting that the input-driven liquid-state update is more effective than static global attention for balancing long-range dependence modeling and local-detail preservation in spectral reconstruction.

3) Physics-Loop Ablation: To further examine the roles of SPC and NDR, we perform a dedicated physics-loop ablation on PaviaU while keeping the front-end (MICSG+BiLSD) unchanged. The following variants are considered: Full Physics-loop (SHP+PEPS+NDR), SHP only, SHP+PEPS (SPC only), SHP+NDR, and No Physics-loop.

As illustrated in Table VIII, SHP alone already improves all four metrics over the No Physics-loop baseline, confirming that explicit projection toward the SRF-feasible set effectively reduces forward inconsistency. Adding PEPS further decreases ERGAS and SAM, indicating that the spectral prior improves spectral plausibility and suppresses unstable null-space oscillations. On the contrary, SHP+NDR tends to improve PSNR and SSIM more than ERGAS and SAM, suggesting that NDR mainly enhances high-frequency details, while PEPS contributes more to global spectral stability. Consequently, the full loop achieves the best overall performance by jointly ensuring forward consistency, spectral-shape stability, and detail refinement.

In addition to the reconstruction metrics, Table VIII further reports the parameter count, FLOPs, and inference time of different physics-loop variants. It can be observed that the additional computational overhead introduced by SHP and PEPS is limited. Compared with the No Physics-loop baseline, SHP only and SHP+PEPS increase the inference time only modestly, indicating that the proposed physics-consistency correction does not introduce a heavy optimization burden. The main complexity increase comes from the NDR refinement branch, which explains why the full physics-loop and SHP+NDR variants are slightly more expensive. Overall, the full physics loop achieves the best reconstruction performance while maintaining a moderate and acceptable computational overhead.

4) Sensitivity Analysis of Subspace Dimension K: To justify the default choice $K = 8$, we conduct a subspace-dimension sensitivity study on the PaviaU dataset. Only the subspace dimension K in MICSG is varied, while all the other settings remain unchanged.

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TABLE VIII
PHYSICS-LOOP ABLATION RESULTS AND RUNTIME ANALYSIS ON THE PAVIAU DATASET

Variant (PaviaU)	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓	Params (M)	FLOPs (G)	Inference time (ms)
Full Physics-loop	39.899	0.9708	6.2064	3.4178	13.5	5.9	28.7
SHP only	39.205	0.9658	6.5489	3.6719	13.0	5.5	23.5
SHP+PEPS (SPC only)	39.521	0.9689	6.3368	3.5069	13.0	5.6	24.7
SHP+NDR	39.412	0.9681	6.4015	3.5642	13.5	5.8	27.4
No Physics-loop	38.846	0.9639	6.9034	3.9881	13.0	5.4	22.8

TABLE IX
SENSITIVITY ANALYSIS OF THE SUBSPACE DIMENSION K ON THE PAVIAU DATASET

K	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓
4	39.214	0.9659	6.6718	3.7816
6	39.643	0.9691	6.3847	3.5472
8	39.899	0.9708	6.2064	3.4178
10	39.841	0.9702	6.2489	3.4526
12	39.706	0.9693	6.3365	3.5184

TABLE X
LOSS FUNCTION WEIGHT ABLATION EXPERIMENT RESULTS ON THE PAVIAU DATASET

Setting ID	λ_c	λ_p	λ_n	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓
Default	0.10	0.50	0.20	**39.899**	**0.9708**	**6.2064**	**3.4178**
C− (↓ 20%)	0.08	0.50	0.20	39.842	0.9701	6.2469	3.4521
C+ (↑ 20%)	0.12	0.50	0.20	39.873	0.9703	6.2328	3.4390
P− (↓ 20%)	0.10	0.40	0.20	39.768	0.9696	6.3187	3.5214
P+ (↑ 20%)	0.10	0.60	0.20	39.855	0.9700	6.2583	3.4636
N− (↓ 20%)	0.10	0.50	0.16	39.928	0.9712	6.2141	3.4448
N+ (↑ 20%)	0.10	0.50	0.24	39.862	0.9704	6.2406	3.4389

As illustrated in Table IX, when K increases from 4 to 8, PSNR and SSIM improve, while ERGAS and SAM decrease, indicating that a moderate increase in K enhances the expressive ability of the coarse spectral skeleton. However, when K is further increased to 10 and 12, the performance no longer improves and instead degrades slightly. This suggests that an excessively large K weakens the low-dimensional constraint and increases the correction burden of the subsequent modules. Therefore, $K = 8$ provides a favorable tradeoff between reconstruction accuracy and model complexity, and is adopted as the default setting.

5) Loss-Weight Sensitivity Ablation: To examine the sensitivity and stability of training with respect to the loss weights, we conduct a small-perturbation study on PaviaU for the three coefficients λ_c, λ_p, and λ_n. The default setting is $(\lambda_c, \lambda_p, \lambda_n) = (0.1, 0.5, 0.2)$. Keeping all other training configurations and the network architecture unchanged, we perturb each coefficient independently by $\pm 20\%$ while fixing the other two at their default values, and report the corresponding trends in PSNR, SSIM, ERGAS, and SAM.

As summarized in Table X, varying the three weights within $\pm 20\%$ does not cause drastic performance fluctuations, indicating that the training objective of PLS^2-Net is reasonably stable and reproducible. Small changes in λ_c (auxiliary supervision for coarse spectra) lead to only minor variations, suggesting that MICSG can learn a stable coarse skeleton under the default setting and the overall performance is not highly sensitive to this term. By contrast, λ_p (SRF forward-consistency loss) has a more visible impact on ERGAS and SAM; the default choice $\lambda_p = 0.5$ provides a balanced tradeoff between projection consistency and reconstruction fidelity. For λ_n, decreasing it slightly increases the flexibility of NDR, which may yield marginal gains in PSNR/SSIM but can also introduce mild projection drift, leading to a slightly worse SAM. Increasing λ_n more strongly constrains the residual toward the null space, improving projection stability but limiting detail compensation, which results in a small drop in PSNR/SSIM.

Overall, $(\lambda_c, \lambda_p, \lambda_n) = (0.1, 0.5, 0.2)$ provides a stable compromise between reconstruction accuracy and physical consistency, and the method remains robust under reasonable perturbations of these weights.

TABLE XI
ABLATION STUDY ON PHYSICAL CONSISTENCY PARAMETERS IN SPC ON THE PAVIAU DATASET

λ (SHP)	μ (PEPS)	η (PEPS)	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓
0.05	1.0	0.01	39.712	0.9701	6.2840	3.4852
0.10	1.0	0.01	**39.899**	**0.9708**	**6.2064**	**3.4178**
0.20	1.0	0.01	39.783	0.9704	6.2556	3.4529
0.10	0.5	0.01	39.664	0.9699	6.3128	3.5036
0.10	1.5	0.01	39.731	0.9702	6.2794	3.4761
0.10	1.0	0.005	39.706	0.9700	6.2912	3.4938
0.10	1.0	0.02	39.681	0.9698	6.3056	3.5104

6) Ablation Study on Physical Consistency Parameters: In addition to the loss-weight ablation, we further investigate the sensitivity of the proposed model to the key physical consistency parameters in the SPC module, including the consistency strength λ in the SHP and the weighting factors μ and η in the PEPS. In particular, λ controls the correction magnitude of the SRF-consistent projection, while μ and η balance the data-consistency term and the spectral prior regularization in PEPS. All other settings are kept unchanged, and each parameter is varied around its default value to evaluate the robustness of the reconstruction performance.

As illustrated in Table XI, the SPC module exhibits robust performance against critical physical consistency parameters, with only minor fluctuations within reasonable perturbation ranges. The SHP achieves optimal balance at $\lambda = 0.1$: values that are too small result in insufficient forward correction, while excessively large values may cause overly strong projections that suppress fine details. In PEPS ($\mu = 1.0$, $\eta = 0.01$), a better compromise is achieved between data consistency and spectral prior constraints; when μ deviates, or η is too strong/weak, both ERGAS and SAM show slight deterioration.

I. Downstream Classification Evaluation

To further assess the practical utility of reconstructed HSI, we conduct a downstream land-cover classification experiment on PaviaU. The spectral features are standardized by

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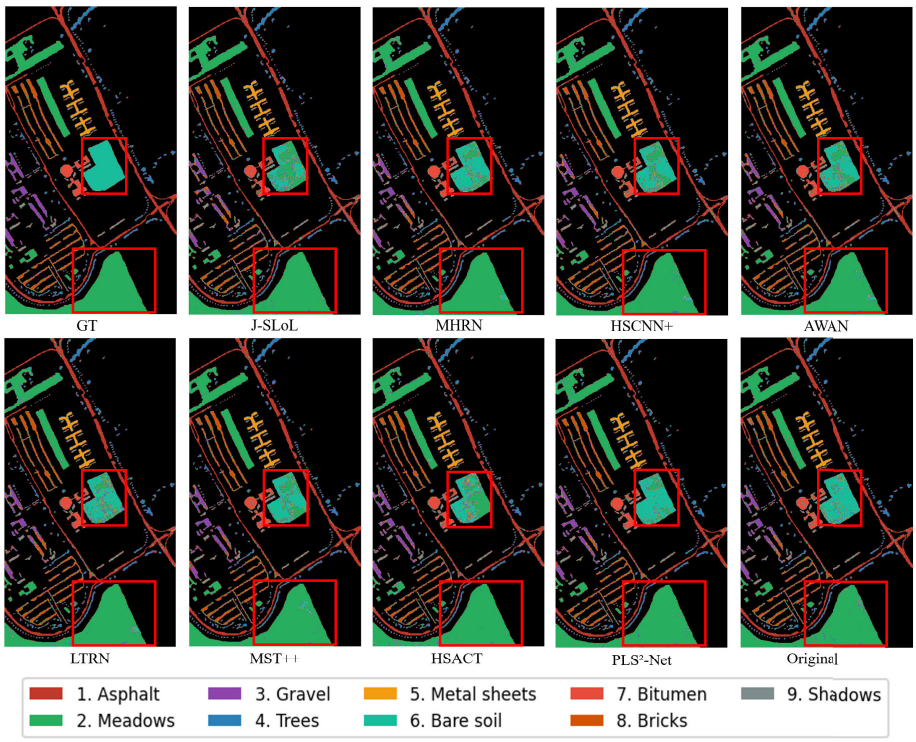


Fig. 9. Comparison of computational complexity and accuracy among methods on the PaviaU dataset.

TABLE XII

DOWNSTREAM CLASSIFICATION RESULTS ON THE PAVIAU DATASET USING HYPERSPECTRAL IMAGES RECONSTRUCTED BY DIFFERENT SSR METHODS

Metric	Hyperspectral images generated by different SSR methods								
	J-SLoL	MHRN	HSCNN+	AWAN	LTRN	MST++	HSACT	PLS^2-Net	Original
OA (%)	82.045	88.511	88.785	91.403	91.367	92.863	92.716	93.956	94.268
AA (%)	80.566	85.326	86.334	89.091	88.736	91.178	91.503	92.103	92.157
Kappa	0.7855	0.8433	0.8482	0.8843	0.8842	0.9044	0.9020	0.9192	0.9265

StandardScaler and reduced from 103 to 30 dimensions using PCA. An SVM with the RBF kernel is adopted, with $C = 100$, $\gamma = \texttt{scale}$, and the one-versus-rest strategy for multiclass classification. OA, AA, and Kappa are reported, and all methods are evaluated under the same preprocessing and classifier settings.

As illustrated in Table XII, PLS^2-Net achieves the best classification performance among all SSR methods, with OA, AA, and Kappa of 93.956%, 92.103%, and 0.9192, respectively. Compared with MST++, PLS^2-Net improves OA, AA, and Kappa by 1.093 percentage points, 0.925 percentage points, and 0.0148. Its results are also very close to those of the original HSI. Fig. 9 further shows that PLS^2-Net produces more spatially coherent predictions and clearer category boundaries. These results indicate that PLS^2-Net preserves more class-discriminative spectral information and offers better utility for downstream remote sensing tasks.

J. Cross-Region Generalization Under Material-Distribution Shifts

To explicitly evaluate cross-region generalization under scene and material-distribution shifts, we conduct bidirectional transfer experiments between PaviaU and Chikusei. These two datasets represent different geographical regions and contain substantially different scene structures, land-cover compositions, and material distributions. Since PLS^2-Net is designed for RGB-to-HSI SSR rather than transfer learning, this experiment is used only as a supplementary evaluation of cross-dataset behavior.

TABLE XIII

CROSS-DATASET GENERALIZATION RESULTS BETWEEN PAVIAU AND CHIKUSEI

Train set → Test set	Method	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓
PaviaU → Chikusei	MST++	36.284	0.9524	18.9621	2.7146
	HSACT	36.941	0.9518	18.2765	2.8014
	PLS^2-Net	37.168	0.9598	17.8465	2.4317
Chikusei → PaviaU	MST++	34.964	0.9328	8.9462	5.6384
	HSACT	34.516	0.9296	8.9738	5.7004
	PLS^2-Net	35.612	0.9412	8.2075	5.1236

The two datasets are first aligned to a common wavelength range, and the Chikusei spectra are resampled to the same 103-band spectral axis as PaviaU. RGB observations are synthesized using the same nominal SRF as in the main experiments. A zero-shot protocol is adopted, where training is performed on the source dataset, and testing is directly conducted on the target dataset without fine-tuning. MST++, HSACT, and PLS^2-Net are compared.

As illustrated in Table XIII, all methods degrade under cross-dataset testing, confirming the increased difficulty caused by distribution shift. Nevertheless, PLS^2-Net still achieves the

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best results in both transfer directions, with more evident advantages in ERGAS and SAM. This suggests that the proposed coarse-to-fine physics-guided framework retains a certain degree of stability under distribution shift, while more dedicated transfer-learning or generalization-enhanced designs will be investigated in future work.

V. Conclusion

Motivated by the severely illposed nature of remote sensing SSR, this article proposes PLS^2-Net, a physics-guided liquid spectral-sequence coarse-to-fine network. The method adopts an interpretable subspace-basis–coefficient-map representation to construct a stable coarse spectral skeleton, and then models long-range cross-band dependencies and directional spectral evolution in the coefficient domain via BiLSDs. In addition, under the SRF-based forward imaging model, PLS^2-Net introduces a differentiable physics-consistent correction loop together with a null-space-constrained detail refinement scheme, which enhances detail compensation while preserving observation-domain consistency. Comprehensive experiments on PaviaU, Chikusei, and Indian Pines demonstrate that the proposed method achieves the best or highly competitive performance in terms of PSNR, SSIM, ERGAS, and SAM, and exhibits more stable error suppression and more consistent spectral reconstruction trends in both RMSE visualizations and spectral profile comparisons.

Despite these improvements, the proposed approach has limitations. In particular, it relies on an accurate and reliable SRF to construct forward-consistency constraints. When the SRF is subject to calibration errors or when SRF mismatch occurs across sensors, the physics-based correction may introduce bias or reduce the expected benefit.

In future work, we plan to investigate SRF uncertainty modeling and adaptive calibration strategies, so that the network can maintain physical consistency and robustness under mild SRF deviations or even partially unknown SRF conditions. We will further investigate generalization-enhanced and transfer-learning strategies for cross-region, cross-scene, and cross-sensor settings, so as to improve the adaptability of the framework under distribution shift. In addition, validation on real paired RGB–HSI data remains an important direction for future work when such accurately aligned benchmark data become available.

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Huijie Jia received the M.S. degree in electronic information from Qilu University of Technology (Shandong Academy of Sciences), Jinan, China, in 2023. He is currently pursuing the Ph.D. degree with the School of Internet of Things Engineering, Jiangnan University, Wuxi, China.

His research interests include spectral image analysis, deep learning, and visible-spectrum detection.



Chenlin Wu received the B.S. degree in building electrical and intelligentization from Suzhou University of Science and Technology, Suzhou, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Internet of Things Engineering, Jiangnan University, Wuxi, China.

His research interests include image analysis and deep learning.



Min Huang received the Ph.D. degree in control science and engineering from Northeastern University, Shenyang, China, in 2004.

She was a Visiting Professor at Michigan State University, East Lansing, MI, USA, from 2008 to 2009, and with the United States Department of Agriculture, Agricultural Research Service, Beltsville, MD, USA, from 2013 to 2014. She is currently a Professor with the Key Laboratory of Advanced Process Control for Light Industry, Ministry of Education, Institute of Automation, Jiangnan University, Wuxi, China. Her research interests include advanced optical inspection technology, complex system modeling, pattern recognition, machine learning, image analysis, deep learning, and Raman spectroscopy.



Qibing Zhu received the Ph.D. degree in mechanical and electronic engineering from Northeastern University, Shenyang, China, in 2006.

From 2012 to 2013, he was a Visiting Professor at Michigan State University, East Lansing, MI, USA. He is currently a Professor with the Key Laboratory of Advanced Process Control for Light Industry, Ministry of Education, Jiangnan University, Wuxi, China. His research is related to the application of deep learning, sensing and detection technology, spectral imaging technology, and sensor integration.

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